The FOMC Window

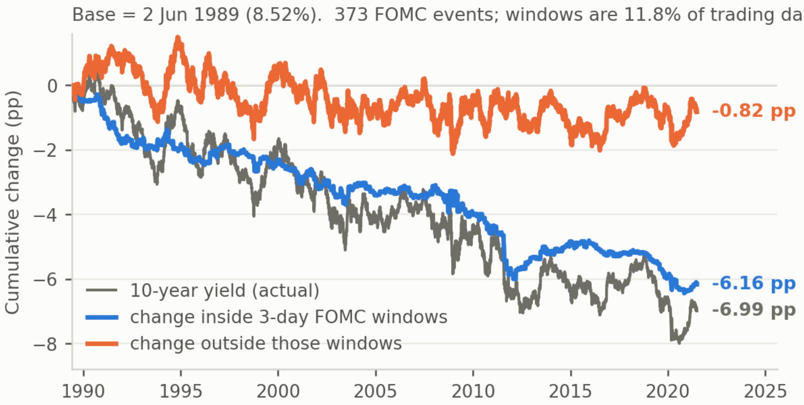
What is identified, what is not, and where the project goes next

Progress report

11 August 2026

The motivating fact

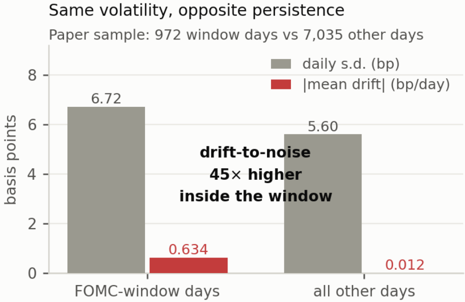
Long rates fall only when the Fed is in the room



Hillenbrand (2025, *RFS*), replicated from GSW yields and a hand-built FOMC calendar.

The fact is really an absence

- Three-day windows are 11.8% of trading days but carry 88% of a 6.99 pp decline.
- The per-meeting effect is small: -2.5 bp.
- What is remarkable is the other 88% of days: drift of -0.012 bp/day, $t = -0.18$, over 7,035 days.
- Randomisation test: 324 randomly placed 3-day blocks on non-FOMC days, 5,000 draws. Observed -6.16 pp sits at the 0th percentile, $p < 0.0002$.



The question this raises

Motivation

If the Fed sets an overnight rate, why does the *ten-year* rate move only on its calendar — and why do those moves not wash out?

The question I set out to answer

When the Fed moves long rates at scheduled announcements, is it changing where markets think rates are going (expected short rates), or what markets charge for the risk of being wrong (the term premium)?

Accounting identity behind it: $y_t^{(n)} = r_t^* + \pi_t^* + \frac{1}{n}\sum_j E_t[\text{gap}_{t+j}] + TP_t^{(n)}$

With the inflation target fixed, "the Fed moves long rates" is close to "the Fed moves beliefs about r^*" — which is why r^* became central to the project. The next slide says how close.

What exactly is the "gap"?

Define the neutral nominal rate $i_t^* \equiv r_t^* + \pi_t^*$ and the *policy gap* $\text{gap}_t \equiv i_t - i_t^*$ — the stance of policy. Under a Taylor rule, $\text{gap}_t = \phi_\pi(\pi_t - \pi_t^*) + \phi_x x_t + v_t$: cyclical and mean-reverting.

If it decays as an AR(1) with monthly persistence ρ, then $\frac{1}{n}\sum_j E_t[\text{gap}_{t+j}] = \omega(\rho, n)\,\text{gap}_t$ with $\omega = \dfrac{1-\rho^n}{n(1-\rho)}$ — small, but not zero:

Half-life of the gap	ω, 2y	ω, 5y	ω, 10y
1 year	0.56	0.29	0.15
2 years	0.73	0.48	0.28
4 years	0.85	0.67	0.48

Over an announcement window,

$$\Delta y_t^{(n)} = \underbrace{\Delta r_t^* + \Delta \pi_t^*}_{\text{endpoint news}} + \underbrace{\omega\,\Delta\text{gap}_t + \text{gap}_t\,\Delta\omega}_{\text{stance and persistence news}} + \Delta TP_t^{(n)}.$$

Four terms, one observable — and π^* is only plausibly fixed after $\sim$2000.

What the literature has established

The core fact and its extensions

Paper	What it establishes
Hillenbrand (2025, *RFS*)	The whole secular decline in the 10-year accumulates in 3-day FOMC windows, 1989–2021. Attributes it to markets learning a lower r^* from the Fed.
Hofmann, Li & Wu (BIS 1252)	The pattern is global — 10 advanced economies — and survives a shifting-endpoint model.
Hanson, Lucca & Wright (2021, *QJE*)	Announcement-window long-rate moves *substantially reverse* within 6–12 months. Read as evidence of temporary term-premium effects.
Bauer & Swanson (2023, *NBER MA*)	Announcement surprises are predictable from pre-meeting public data; the "Fed information effect" is largely the Fed responding to news.

Rows 1 and 3 are in tension: Hillenbrand's fact requires window moves *not* to reverse.

The contested channel

Paper	What it establishes
Hofmann, Li & Wu (BIS 1252)	The window decline is risk-neutral (US pseudo-R^2 96 vs 32). But the endpoint is defined as PC1 of cumulative FOMC-window yield changes — it inherits the series being explained.
Pan & Peng (SSRN 4764451)	On day $t-1$ the decline is term premium: -0.71 of -0.79 bp; expectations -0.08, insignificant.
Joslin, Singleton & Zhu (2011); Joslin, Priebsch & Singleton (2014)	Under spanning the state is recoverable from yields, so the expectations component is a fixed linear function of the curve.
Bauer & Rudebusch (2020, *AER*); (2014, *IJCB*)	A stationary VAR books genuine endpoint shifts as term premium: 75% vs 35–37% for the secular decline.

Two papers, opposite answers, neither adjudicated.

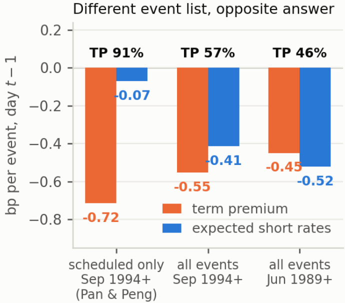
The beliefs-about-r^* strand

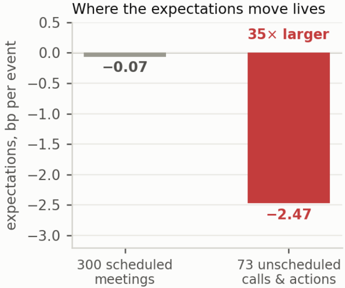
Paper	**What it establishes**
Bauer, Pflueger & Sunderam (2024, *QJE*)	A pre-meeting *subjective belief* — the perceived policy reaction-function *slope* — is related to term premia and to state-dependent window responses. The template for this design.
Ahonon & Roussellet (FRBNY SR 1187)	Under learning about unobserved trends, investor uncertainty about r^* is ± 170 bp, not ± 55 bp; r^* is the primary long-run driver of bond risk premia.
Couture (2021); Engstrom (FEDS 2026-026)	The dot plot moves asset prices; SEP-minus-market gaps predict forecast errors.
Favero, Melone & Tamoni (2024, *JFQA*)	A policy-rate-minus-equilibrium-rate gap predicts bond returns. Different wedge, similar name — must be differentiated explicitly.

Never tested: the perceived *endpoint* — not how strongly markets think the Fed responds, but where they think it is heading.

What our extensions add

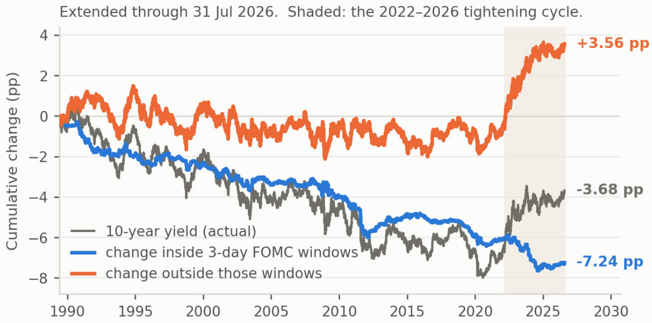
Extension 1: the event list decides the answer





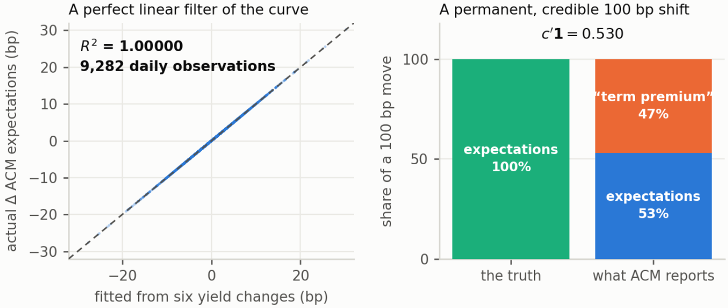
- Pan & Peng replicated to three coefficients (−0.785 / −0.716 / −0.070). Their filter — scheduled meetings only, 1994+ — drops 73 calls and intermeeting actions.
- Same day, same data, same decomposition: the term-premium share moves 91% → 46%.
- **New:** the two contradictory papers are reconciled without taking a side on the model.

Extension 2: the sample through July 2026



- 2022–2026: the Fed raised the target 525 bp and the 10-year rose 2.60 pp. In windows it *fell* 1.20 pp — the entire increase happened outside.
- The in-window series keeps declining, to −7.24 pp, straight through the tightening.
- **New:** no paper covers this period, and the pattern is not symmetric in the direction of policy.

Extension 3: the decomposition cannot arbitrate



- $\Delta rn_t^{(n)} = c_n' \Delta y_t^{\mathcal{N}}$, $c_n' = b_n' \mathrm{B}^{-1}$; empirically $R^2 = 1.00000$, so $\beta^{rn} = c'\beta^{\mathcal{N}}$ (predicted $+2.5/+29.3$ vs estimated $+2.6/+29.2$ bp).
- The decomposition adds no event-specific information, and $c'1 = 0.529$ mis-books half of a permanent credible shift.
- **Attribution:** algebra from JSZ (2011), bias from Bauer–Rudebusch; mine is the ACM filter and the event-study corollary.

Where r^* enters

Why r^* is the natural mechanism — and the natural test

The logic

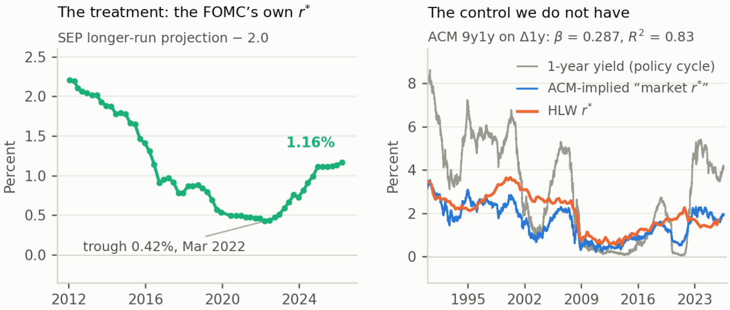
The Fed's 2% target is fixed and credible, so a permanent move in the 10-year that is *not* a risk premium has to be a move in beliefs about the real endpoint. "The Fed moves long rates" $\Rightarrow$ "the Fed moves beliefs about r^*."

The design it suggested

Does the pre-meeting gap between the Fed's published longer-run projection and the market-implied endpoint predict the window response — and does it load on the term premium or on expectations?

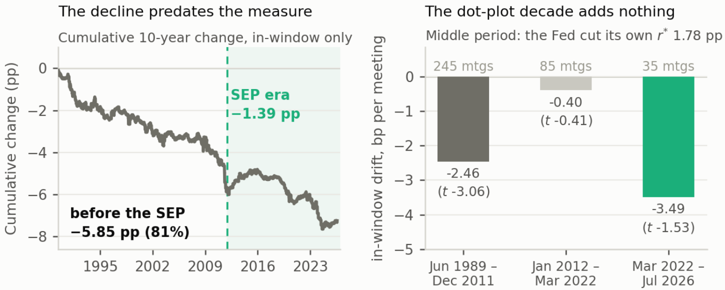
- Needs a Fed leg: the SEP longer-run dot, available from 2012.
- Needs a market leg: an endpoint measure *not* built from FOMC-window data.

The treatment exists; the control does not



- The Fed leg is real and moves: −1.78 pp to Mar 2022, then +0.74 pp; 48 distinct values in the mean dot.
- The market leg fails — the ACM 9y1y forward is a damped copy of the policy cycle ($\beta = 0.287$, $R^2 = 0.83$, level corr. +0.984).
- Structural, not a bad horizon choice: under a stationary VAR $b_n \to 0$, so *no* horizon isolates an endpoint. Survey alternatives correlate −0.12 with it.

And the treatment window is the wrong decade



- 81% of the in-window decline is complete before the dot plot begins.
- In the decade when the Fed cut its own longer-run rate 1.78 pp — when an information channel should be loudest — the in-window drift is -0.40 bp/meeting, $t = -0.41$.
- Every projection-based test of this channel, the published one included, is run where there is almost nothing left to explain.

Preliminary results and next steps

What survives

Result	Identifying assumption	Status
Event-list reconciliation: TP share 91% $\rightarrow$ 46%	None beyond the ACM series being the same object in both papers	Finished; most immediately publishable
ACM is a fixed filter, $R^2 = 1$, $c'1 = 0.529$	Spanning; ACM's own estimated loadings	Finished; theorem standard, application new
Timing fact: 81% pre-dates the SEP	Window assignment only	Finished; reframes the projection literature
2022–26: tightening occurs entirely outside windows	Window assignment only	Finished; no other paper covers it
Fed–market wedge design	Requires a non-circular market endpoint	Set aside — no such measure available

The unanswered question I would build on

The puzzle

Window and non-window days have nearly identical daily volatility (6.72 vs 5.60 bp) but completely different cumulative outcomes (-6.16 vs -0.82 pp). Drift-to-noise is $45\times$ higher inside the window.

So the question is not *which component moves* — it is

why do only scheduled-date price changes persist?

- A long-horizon variance-ratio decomposition by day type turns this from a description into a measured object.
- It needs no decomposition, no endpoint measure, and no model of the term premium — so it survives every identification problem above.
- It confronts Hanson–Lucca–Wright (reversal) with Hillenbrand (permanence) directly.

Path forward

Immediate (four weeks)

(a) **Issuance test.** Lou, Pintér, Üslü & Walker (BIS 1232) claim the effect concentrates in windows *without* long-dated issuance. Auction dates are public; decisive either way in days.

(b) **Draft the reconciliation note** (event list + timing fact). Self-contained and model-free.

Medium term (one term)

(c) **Permanent vs transitory by day type** — the variance-ratio design above. This is the chapter.

Contingent

(d) **Extend the Fed-side measure back before 2012** using Tealbook/Greenbook projections (1981–2019, five-year lag) — the only way to test any projection-based mechanism on the sample that contains the phenomenon.

Where I would value guidance

1. Is the reconciliation better as a standalone short submission, or folded into a chapter as a methodological section?
2. Access to Blue Chip Financial Forecasts long-range and to the Tealbook archive — both bind on (d).
3. Is the permanent/transitory framing in (c) the right way to pose the question, or too descriptive to carry a chapter without a mechanism?

Data. Gürkaynak–Sack–Wright zero-coupon yields; ACM daily decomposition, Jun 1989 – Aug 2026; FOMC calendar built from Federal Reserve historical meeting pages (300 scheduled meetings plus 73 conference calls and unscheduled actions); SEP dot-plot archive, 57 meetings, Jan 2012 – Mar 2026. All estimates recomputed from source.